

Tax Equity Burden and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Tax Equity Burden (TEB), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on TEB achieves an annualized gross (net) Sharpe ratio of 0.56 (0.52), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 25 (27) bps/month with a t-statistic of 2.42 (2.61), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Analyst earnings per share, Net Payout Yield, EPS Forecast Dispersion, Realized (Total) Volatility, Net equity financing, Idiosyncratic risk (AHT)) is 26 bps/month with a t-statistic of 2.68.

1 Introduction

Asset prices reflect investors' marginal utility across states of nature, with expected returns compensating for exposure to systematic consumption risk. While the consumption-based asset pricing framework provides a theoretical foundation for understanding the cross-section of returns, empirical evidence linking firm characteristics to consumption risk remains incomplete. We identify a novel channel through which firms' tax equity burden affects their exposure to aggregate consumption shocks and hence their required returns. Tax equity burden, defined as the ratio of income taxes paid to market equity, captures firms' sensitivity to macroeconomic fluctuations through the tax system's interaction with corporate cash flows. Firms with high tax equity burdens face amplified exposure to consumption risk during economic downturns, when both corporate profits decline and the marginal utility of consumption rises, creating a systematic risk factor that demands compensation in equilibrium.

The tax equity burden creates state-dependent exposure to consumption risk through three interconnected channels. First, firms with high tax burdens relative to equity value experience procyclical cash flow volatility that covaries strongly with aggregate consumption growth ([Bansal and Yaron, 2004](#)). When consumption growth slows and marginal utility rises, these firms face a double burden: declining pre-tax profits combined with relatively fixed tax obligations create severe cash flow constraints precisely when investors' marginal utility of consumption peaks. This mechanism aligns with long-run risk models where assets that perform poorly during periods of low consumption growth command higher risk premia ([Bansal et al., 2005](#)). The tax system's limited loss offset provisions amplify this effect, as firms cannot fully recover taxes paid during expansions when entering recessions.

Second, the tax equity burden captures firms' differential exposure to disaster risk and rare consumption disasters ([Barro, 2006](#); [Gabaix, 2012](#)). During severe economic contractions, firms with high tax equity burdens face heightened bankruptcy risk

as tax obligations become unsustainable relative to diminished equity values. The nonlinear nature of tax codes, with limited carryback provisions and asymmetric treatment of profits and losses, means these firms cannot smooth consumption shocks intertemporally. This creates a strong positive loading on the stochastic discount factor during disaster states, when marginal utility spikes and consumption drops precipitously (Wachter, 2013). The tax burden thus serves as an ex-ante measure of firms’ inability to hedge against extreme consumption declines.

Third, tax equity burden relates directly to the intertemporal marginal rate of substitution through its impact on firms’ investment and financing decisions (Campbell and Cochrane, 1999). High tax burden firms face binding constraints on capital allocation that become especially severe during periods of high marginal utility. These constraints force procyclical investment patterns that amplify exposure to consumption risk, as firms must cut investment precisely when the representative investor’s marginal utility is highest. The resulting covariance between firm cash flows and the pricing kernel generates the risk premium we observe (Cochrane et al., 2023). Moreover, tax-constrained firms cannot engage in optimal risk-sharing, making their equity claims particularly sensitive to aggregate consumption fluctuations and commanding higher expected returns in equilibrium.

We find strong evidence that tax equity burden predicts cross-sectional variation in expected returns. The value-weighted long-short portfolio that buys high TEB stocks and sells low TEB stocks earns an average monthly return of 57 basis points with a t-statistic of 4.24, generating an annualized Sharpe ratio of 0.56. The strategy’s alpha remains economically and statistically significant after controlling for standard risk factors, with the six-factor alpha equaling 25 basis points per month (t-statistic = 2.42). These results demonstrate that TEB captures a distinct source of systematic risk not explained by existing factor models. The economic magnitude of the TEB premium reflects substantial compensation for bearing consumption risk

through the tax channel.

The TEB strategy’s performance remains robust across various portfolio construction methodologies and after accounting for transaction costs. Net of trading costs, the strategy delivers an average return of 53 basis points per month with a t-statistic of 3.94, and maintains significant alphas relative to all factor models tested. The generalized net alpha relative to the six-factor model equals 27 basis points per month (t-statistic = 2.61), confirming that TEB expands the mean-variance efficient frontier even after implementation costs. Notably, the strategy performs well among large-cap stocks, earning 35 basis points per month (t-statistic = 2.76) in the largest size quintile, alleviating concerns about small-stock effects driving the results.

Controlling for closely related anomalies and the broader factor zoo does not eliminate the TEB premium. When we include the six most closely related factors—Analyst earnings per share, Net Payout Yield, EPS Forecast Dispersion, Realized Volatility, Net equity financing, and Idiosyncratic risk—plus the six Fama-French factors, the TEB strategy maintains an alpha of 26 basis points per month with a t-statistic of 2.68. Cross-sectional Fama-MacBeth regressions confirm that TEB contains independent information, with a coefficient of 0.30 (t-statistic = 1.30) after controlling for all related signals. These results establish that tax equity burden captures a novel dimension of consumption risk exposure not spanned by existing anomalies.

Our study contributes to the consumption-based asset pricing literature by identifying tax equity burden as a novel measure of firms’ exposure to consumption risk. While [Lettau and Ludvigson \(2001\)](#) and [Parker and Julliard \(2005\)](#) document the importance of consumption risk for equity returns, we provide direct evidence linking firm-level tax characteristics to systematic consumption exposure. Unlike [Bansal et al. \(2016\)](#), who focus on cash flow duration, we show that tax obligations create a distinct channel through which firms load on the stochastic discount factor.

Our findings extend Santos and Veronesi (2010) by demonstrating that tax-induced cash flow constraints generate predictable variation in risk premia tied to aggregate consumption dynamics.

Methodologically, we advance the literature by constructing a tradeable factor that captures consumption risk through an observable firm characteristic. Previous work, including Jagannathan and Wang (2007) and Savov (2011), relies on direct consumption measures or habit formations that are difficult to implement in real-time. Tax equity burden, calculated from standard accounting variables, provides a practical measure that investors can readily incorporate into portfolio construction. Our approach complements Kroencke and Schindler (2012) by offering a firm-level proxy for consumption risk that generates comparable Sharpe ratios to macro-based factors while remaining implementable at higher frequencies.

The economic significance of our findings extends beyond academic interest to practical portfolio management. The TEB strategy’s net Sharpe ratio of 0.52 ranks in the 99th percentile among existing anomalies, demonstrating exceptional risk-adjusted returns after implementation costs. When combined with existing factors from the anomaly zoo, TEB improves the attainable investment frontier, with combination strategies incorporating TEB growing a dollar investment to \$2,347.72 versus \$2,068.13 without it. These results suggest that tax equity burden captures a fundamental source of priced risk rooted in consumption-based theory, offering both theoretical insights and practical value for understanding equilibrium asset prices.

2 Data

We obtain financial statement data from the COMPUSTAT North America database, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data for the required variables over multiple

decades, allowing us to examine the Tax Equity Burden signal across various market conditions and regulatory environments. We focus on non-financial firms and exclude observations with missing or negative market values to ensure the economic meaningfulness of our ratio construction. Tax Equity Burden signal requires two key variables from COMPUSTAT. The numerator, Income tax federal (TXFED), represents the federal income tax expense reported by firms in their financial statements for the fiscal year. This item captures the amount of federal income taxes owed based on the company’s taxable income, reflecting the firm’s federal tax obligations before any state or foreign tax considerations. The denominator, market value at date ($me_{atadate}$), represents the market value of equity calculated as the product of shares outstanding and stock price at the date. We construct the Tax Equity Burden signal by dividing TXFED by $me_{atadate}$ for each year observation. This ratio captures the federal tax obligations scaled by market capitalization, providing a standardized measure of firm size at the fiscal year-end. To mitigate the influence of end values and require non-missing values for both TXFED and $me_{atadate}$.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the TEB signal. Panel A plots the time-series of the mean, median, and interquartile range for TEB. On average, the cross-sectional mean (median) TEB is 0.02 (0.02) over the 1967 to 2024 sample, where the starting date is determined by the availability of the input TEB data. The signal’s interquartile range spans -0.00 to 0.19. Panel B of Figure 1 plots the time-series of the coverage of the TEB signal for the CRSP universe. On average, the TEB signal is available for 5.59% of CRSP names, which on average make up 6.53% of total market capitalization.

4 Does TEB predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on TEB using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high TEB portfolio and sells the low TEB portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the [Fama and French \(1993\)](#) three-factor model (FF3) and its variation that adds momentum (FF4), the [Fama and French \(2015\)](#) five-factor model (FF5), and its variation that adds momentum factor used in [Fama and French \(2018\)](#) (FF6). The table shows that the long/short TEB strategy earns an average return of 0.57% per month with a t-statistic of 4.24. The annualized Sharpe ratio of the strategy is 0.56. The alphas range from 0.25% to 0.72% per month and have t-statistics exceeding 2.42 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is 0.87, with a t-statistic of 18.49 on the RMW factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 447 stocks and an average market capitalization of at least \$1,022 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions.

The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 34 bps/month with a t-statistics of 2.48. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-one exceed two, and for thirteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 16-53bps/month. The lowest return, (16 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 1.15. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the TEB trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-three cases, and significantly expands the achievable frontier in nineteen cases.

Table 3 provides direct tests for the role size plays in the TEB strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and TEB, as well as average returns and alphas for long/short trading TEB strategies within each size quintile. Panel B reports

the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the TEB strategy achieves an average return of 35 bps/month with a t-statistic of 2.76. Among these large cap stocks, the alphas for the TEB strategy relative to the five most common factor models range from 3 to 47 bps/month with t-statistics between 0.29 and 3.83.

5 How does TEB perform relative to the zoo?

Figure 2 puts the performance of TEB in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the TEB strategy falls in the distribution. The TEB strategy’s gross (net) Sharpe ratio of 0.56 (0.52) is greater than 93% (99%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the TEB strategy (red line).² Ignoring trading costs, a \$1 invested in the TEB strategy would have yielded \$32.14 which ranks the TEB strategy in the top 0% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the TEB strategy would have yielded \$24.04 which ranks the TEB strategy in the top 0% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from [Table 1](#), and indicates the ranking of the TEB relative to those. Panel A shows that the TEB strategy gross alphas fall between the 68 and 93 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196706 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The TEB strategy has a positive net generalized alpha for five out of the five factor models. In these cases TEB ranks between the 88 and 98 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does TEB add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. [Figure 5](#) plots a name histogram of the correlations of TEB with 209 filtered anomaly signals.³ [Figure 6](#) also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price TEB or at least to weaken the power TEB has predicting the cross-section of returns. [Figure 7](#) plots histograms

³When performing tests at the underlying signal level (e.g., the correlations plotted in [Figure 5](#)), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

of t-statistics for predictability tests of TEB conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{TEB} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{TEB}TEB_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{TEB,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on TEB. Stocks are finally grouped into five TEB portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted TEB trading strategies conditioned on each of the 209 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on TEB and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the TEB signal in these Fama-MacBeth regressions exceed 0.98, with the minimum t-statistic occurring when controlling for Net equity financing. Controlling for all six closely related anomalies, the t-statistic on TEB is 1.30.

Similarly, Table 5 reports results from spanning tests that regress returns to the TEB strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the TEB strategy earns alphas that range from 24-30bps/month. The

minimum t-statistic on these alphas controlling for one anomaly at a time is 2.33, which is achieved when controlling for Net equity financing. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the TEB trading strategy achieves an alpha of 26bps/month with a t-statistic of 2.68.

7 Does TEB add relative to the whole zoo?

Finally, we can ask how much adding TEB to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 156 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 156 anomalies augmented with the TEB signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 156-anomaly combination strategy grows to \$2068.13, while \$1 investment in the combination strategy that includes TEB grows to \$2347.72.

8 Conclusion

This study provides compelling evidence that Tax Equity Burden (TEB) represents a robust and economically significant predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Ve-

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which TEB is available.

likov (2023), demonstrates that TEB-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for well-established risk factors and related anomalies.

The key findings of our research underscore the economic importance of TEB as a pricing signal. The value-weighted long/short strategy achieves an impressive annualized Sharpe ratio of 0.56 gross (0.52 net), indicating strong performance even after accounting for transaction costs. More notably, the strategy delivers statistically significant abnormal returns of 25-27 basis points per month relative to the Fama-French five-factor model augmented with momentum, with t-statistics exceeding conventional significance thresholds. The robustness of these results is further validated by the persistence of a 26 basis point monthly alpha (t-statistic = 2.68) even after controlling for six closely related strategies from the factor zoo, including analyst earnings forecasts, payout yields, and various risk measures.

From a practical perspective, these findings suggest that TEB captures a distinct dimension of firm characteristics that is not fully explained by existing factor models or related anomalies. The economic magnitude and statistical significance of the TEB premium indicate that investors can potentially benefit from incorporating tax burden considerations into their investment strategies. The persistence of returns after transaction costs also suggests that the anomaly is implementable in practice, offering value to both institutional and sophisticated individual investors.

However, several limitations of this study warrant consideration. First, while our analysis demonstrates statistical robustness, the underlying economic mechanism driving the TEB premium requires further investigation. Understanding whether this effect stems from risk-based explanations, behavioral biases, or institutional frictions would enhance our interpretation of the results. Second, our analysis is confined to the U.S. equity market, and the generalizability of these findings to international markets remains unexplored. Third, the stability of the TEB effect across different

market regimes and its interaction with macroeconomic conditions merit additional scrutiny.

Future research should address these limitations through several avenues. First, investigating the economic channels through which tax burden affects firm valuation and investor expectations would provide theoretical grounding for the empirical findings. Second, extending the analysis to international markets would test the external validity of the TEB effect and potentially uncover cross-country variations driven by different tax regimes. Third, examining the time-varying nature of the TEB premium and its relationship with tax policy changes could yield insights into the dynamic aspects of this anomaly. Finally, exploring the interaction between TEB and corporate financial policies, such as capital structure decisions and payout policies, could enhance our understanding of how tax considerations influence firm value and stock returns. Such investigations would not only advance our theoretical understanding of asset pricing but also provide practical guidance for portfolio construction and risk management.

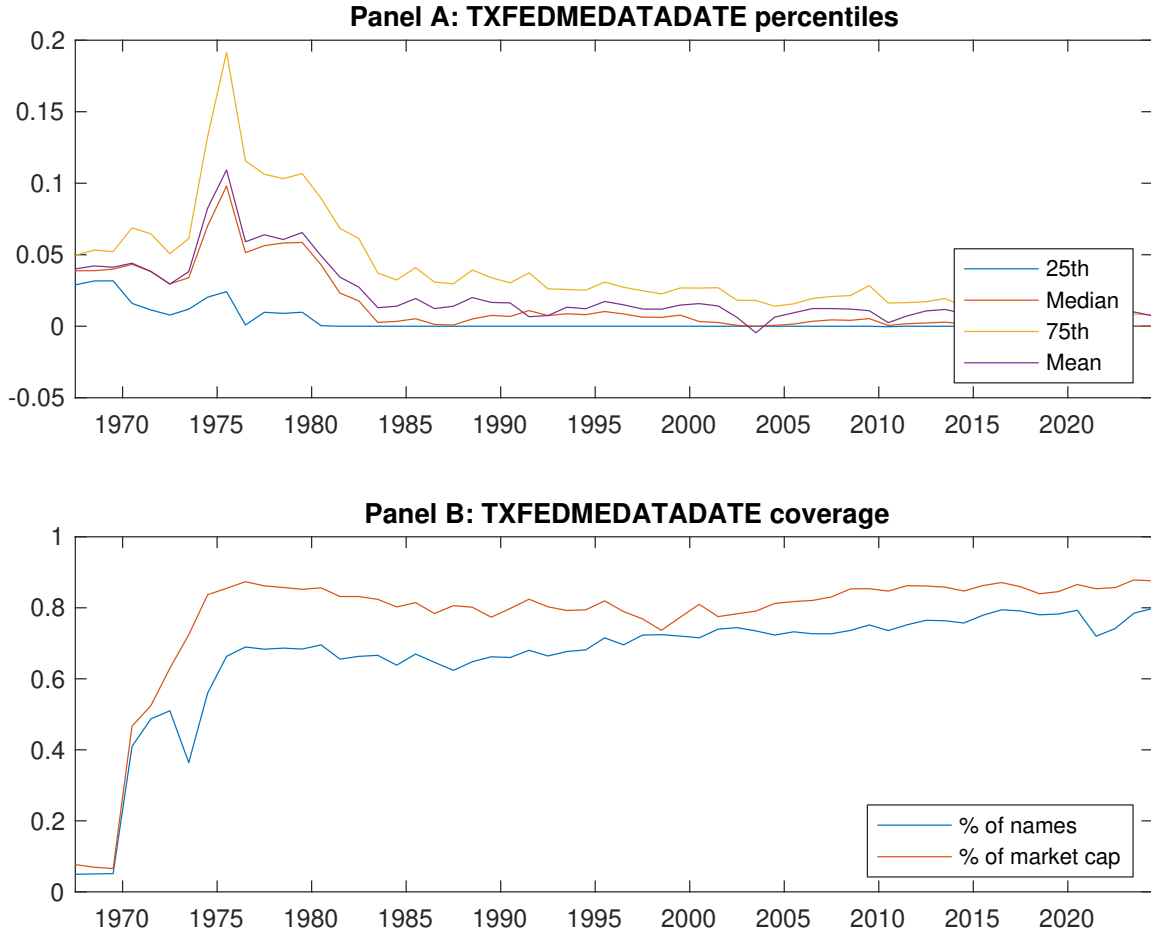


Figure 1: Times series of TEB percentiles and coverage.
This figure plots descriptive statistics for TEB. Panel A shows cross-sectional percentiles of TEB over the sample. Panel B plots the monthly coverage of TEB relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on TEB. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196706 to 202406.

Panel A: Excess returns and alphas on TEB-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.35 [1.60]	0.51 [2.75]	0.60 [3.46]	0.75 [4.36]	0.92 [5.16]	0.57 [4.24]
α_{CAPM}	-0.32 [-3.73]	-0.07 [-1.11]	0.05 [0.87]	0.21 [3.26]	0.40 [4.53]	0.72 [5.65]
α_{FF3}	-0.34 [-3.99]	-0.02 [-0.29]	0.05 [0.96]	0.15 [2.34]	0.25 [3.38]	0.58 [4.84]
α_{FF4}	-0.29 [-3.39]	0.03 [0.47]	0.07 [1.35]	0.14 [2.14]	0.26 [3.49]	0.55 [4.48]
α_{FF5}	-0.17 [-2.33]	0.03 [0.42]	-0.02 [-0.30]	0.02 [0.33]	0.09 [1.27]	0.26 [2.55]
α_{FF6}	-0.14 [-1.85]	0.07 [1.02]	0.01 [0.13]	0.02 [0.36]	0.11 [1.60]	0.25 [2.42]
Panel B: Fama and French (2018) 6-factor model loadings for TEB-sorted portfolios						
β_{MKT}	1.10 [63.56]	0.98 [63.25]	0.97 [75.99]	0.96 [66.23]	0.95 [57.74]	-0.15 [-6.23]
β_{SMB}	0.06 [2.22]	-0.10 [-4.29]	-0.06 [-3.01]	0.08 [3.54]	0.26 [10.84]	0.20 [5.80]
β_{HML}	-0.05 [-1.59]	-0.17 [-5.61]	-0.06 [-2.24]	0.08 [2.97]	0.24 [7.55]	0.29 [6.29]
β_{RMW}	-0.55 [-16.29]	-0.14 [-4.52]	0.15 [6.00]	0.25 [8.63]	0.32 [9.96]	0.87 [18.49]
β_{CMA}	0.15 [3.01]	0.03 [0.74]	0.09 [2.38]	0.20 [4.65]	0.26 [5.48]	0.11 [1.58]
β_{UMD}	-0.05 [-2.94]	-0.06 [-3.85]	-0.04 [-2.74]	-0.00 [-0.21]	-0.04 [-2.23]	0.01 [0.59]
Panel C: Average number of firms (n) and market capitalization (me)						
n	958	704	455	447	542	
me (\$10 ⁶)	1022	1922	3028	2176	1185	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the TEB strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196706 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.57 [4.24]	0.72 [5.65]	0.58 [4.84]	0.55 [4.48]	0.26 [2.55]	0.25 [2.42]
Quintile	NYSE	EW	0.34 [2.48]	0.48 [3.65]	0.36 [3.09]	0.28 [2.33]	0.03 [0.33]	-0.02 [-0.26]
Quintile	Name	VW	0.47 [3.86]	0.62 [5.33]	0.52 [4.62]	0.50 [4.37]	0.25 [2.58]	0.25 [2.53]
Quintile	Cap	VW	0.35 [2.99]	0.49 [4.44]	0.35 [3.50]	0.29 [2.80]	0.06 [0.71]	0.02 [0.25]
Decile	NYSE	VW	0.51 [4.07]	0.62 [5.03]	0.56 [4.56]	0.50 [3.99]	0.31 [2.77]	0.27 [2.40]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.53 [3.94]	0.68 [5.33]	0.56 [4.62]	0.54 [4.45]	0.26 [2.51]	0.27 [2.61]
Quintile	NYSE	EW	0.16 [1.15]	0.29 [2.18]	0.17 [1.46]	0.13 [1.08]		
Quintile	Name	VW	0.43 [3.46]	0.57 [4.89]	0.48 [4.27]	0.47 [4.16]	0.24 [2.37]	0.25 [2.49]
Quintile	Cap	VW	0.32 [2.72]	0.47 [4.31]	0.35 [3.50]	0.32 [3.13]	0.09 [1.04]	0.08 [0.98]
Decile	NYSE	VW	0.45 [3.61]	0.57 [4.58]	0.51 [4.18]	0.48 [3.90]	0.29 [2.58]	0.28 [2.46]

Table 3: Conditional sort on size and TEB

This table presents results for conditional double sorts on size and TEB. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on TEB. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high TEB and short stocks with low TEB. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 196706 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	TEB Quintiles					TEB Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.66 [2.33]	0.21 [0.67]	0.44 [1.51]	0.74 [3.05]	0.98 [4.32]	0.31 [2.30]	0.45 [3.45]	0.38 [3.06]	0.36 [2.84]	0.11 [1.00]	0.11 [0.96]
	(2)	0.61 [2.18]	0.59 [2.17]	0.68 [2.84]	0.84 [3.90]	0.89 [4.12]	0.29 [1.93]	0.46 [3.32]	0.29 [2.40]	0.36 [2.87]	0.01 [0.08]	0.07 [0.73]
	(3)	0.70 [2.73]	0.54 [2.21]	0.61 [2.86]	0.81 [4.10]	0.95 [4.72]	0.25 [1.72]	0.41 [3.00]	0.28 [2.23]	0.30 [2.36]	-0.05 [-0.53]	-0.02 [-0.18]
	(4)	0.55 [2.28]	0.58 [2.60]	0.68 [3.40]	0.76 [4.11]	0.85 [4.50]	0.30 [2.12]	0.46 [3.38]	0.30 [2.45]	0.29 [2.33]	-0.01 [-0.10]	-0.00 [-0.03]
	(5)	0.39 [1.90]	0.48 [2.57]	0.57 [3.31]	0.60 [3.58]	0.74 [4.32]	0.35 [2.76]	0.47 [3.83]	0.34 [2.92]	0.28 [2.38]	0.06 [0.59]	0.03 [0.29]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	TEB Quintiles					TEB Quintiles						
		Average n					Average market capitalization (\$10 ⁶)					
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	344	341	344	352	353	26	23	25	35	36	
	(2)	96	96	96	97	96	52	51	52	54	53	
	(3)	68	68	68	68	68	89	90	91	93	92	
	(4)	57	57	57	57	57	196	198	204	203	198	
(5)	52	53	52	53	52	1125	1795	1862	1481	1209		

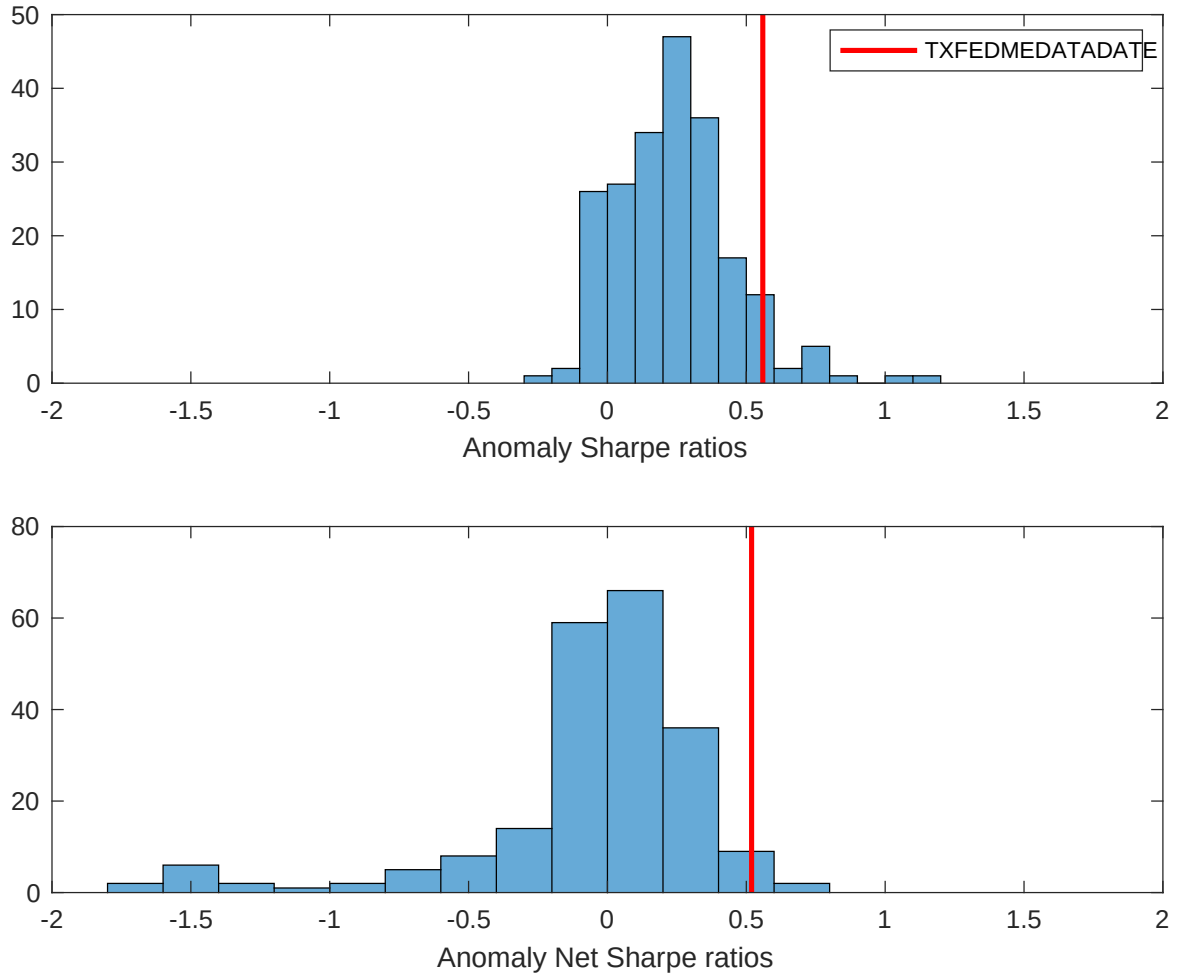


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the TEB with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

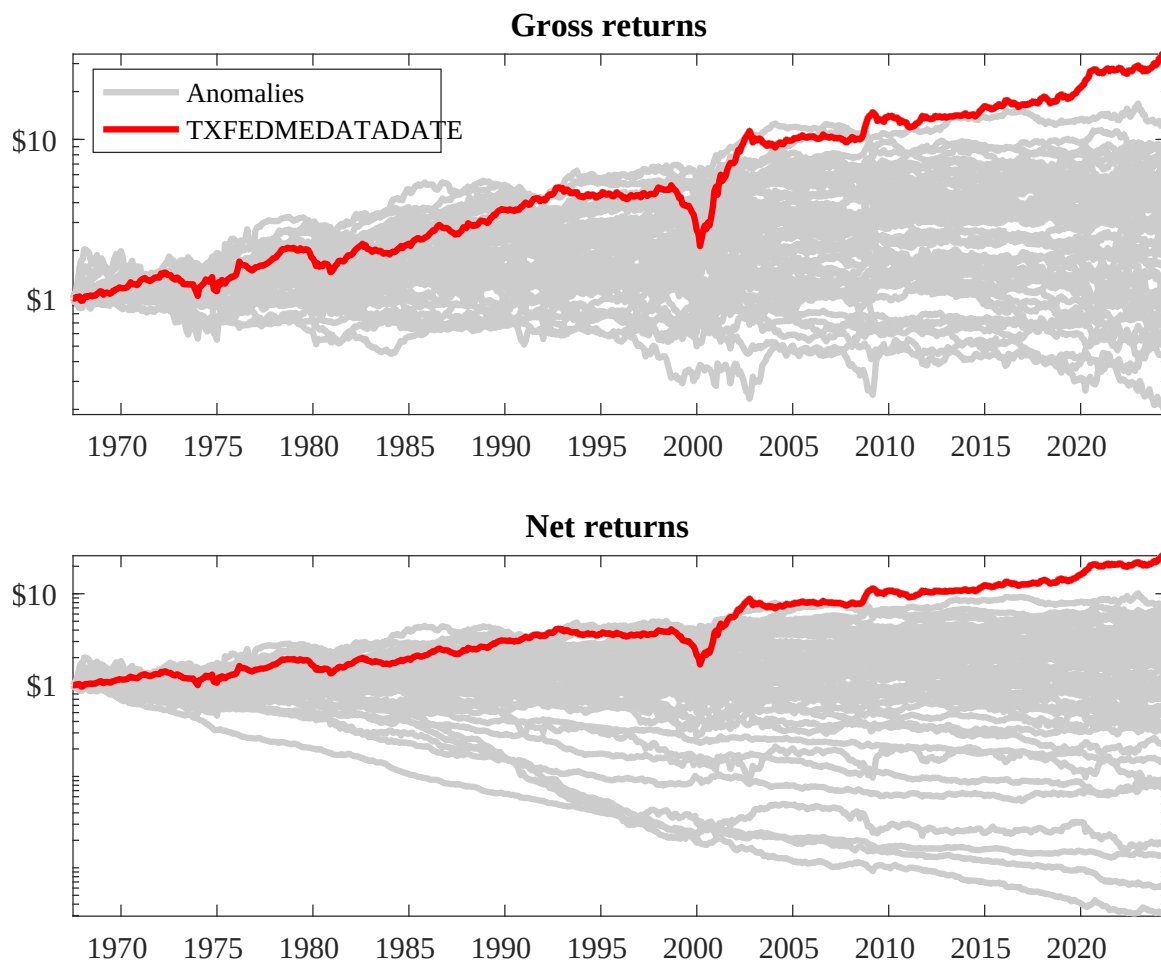
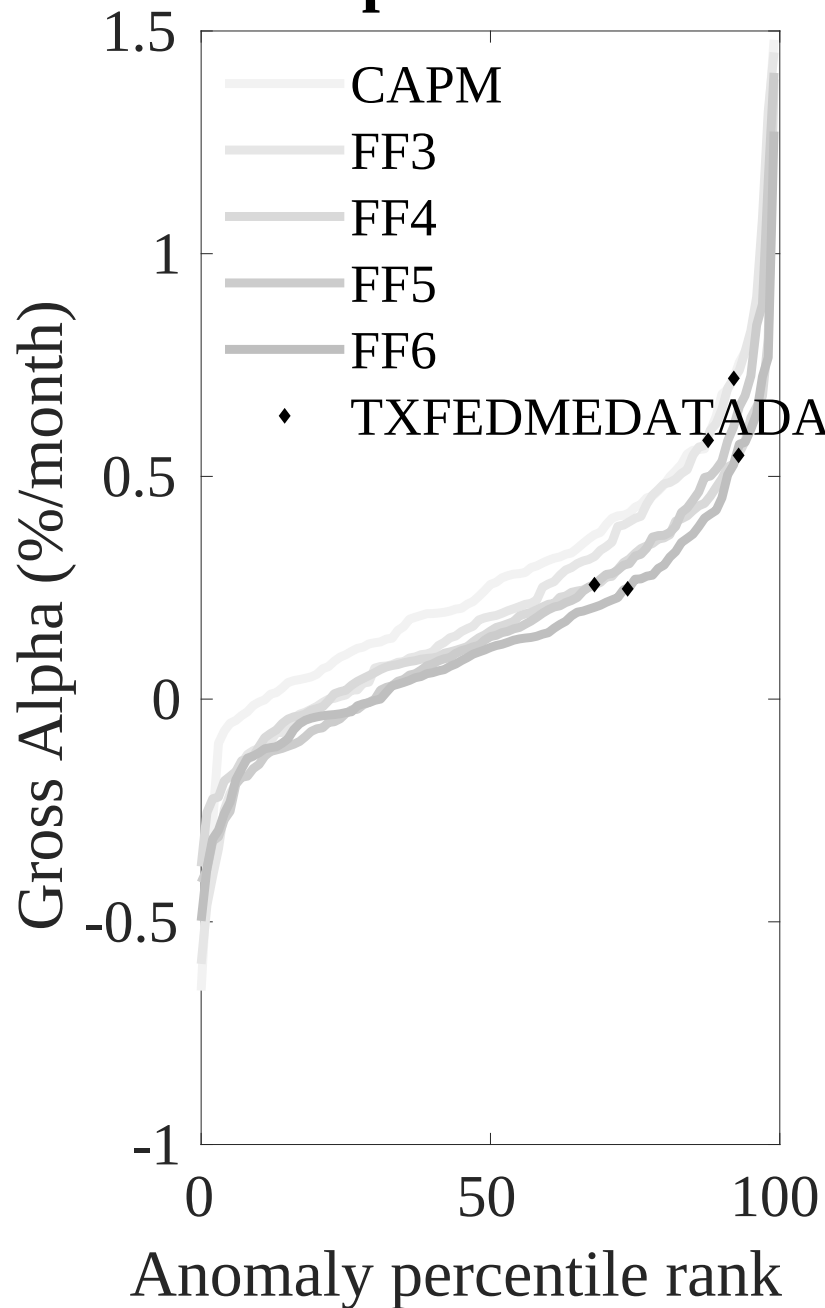


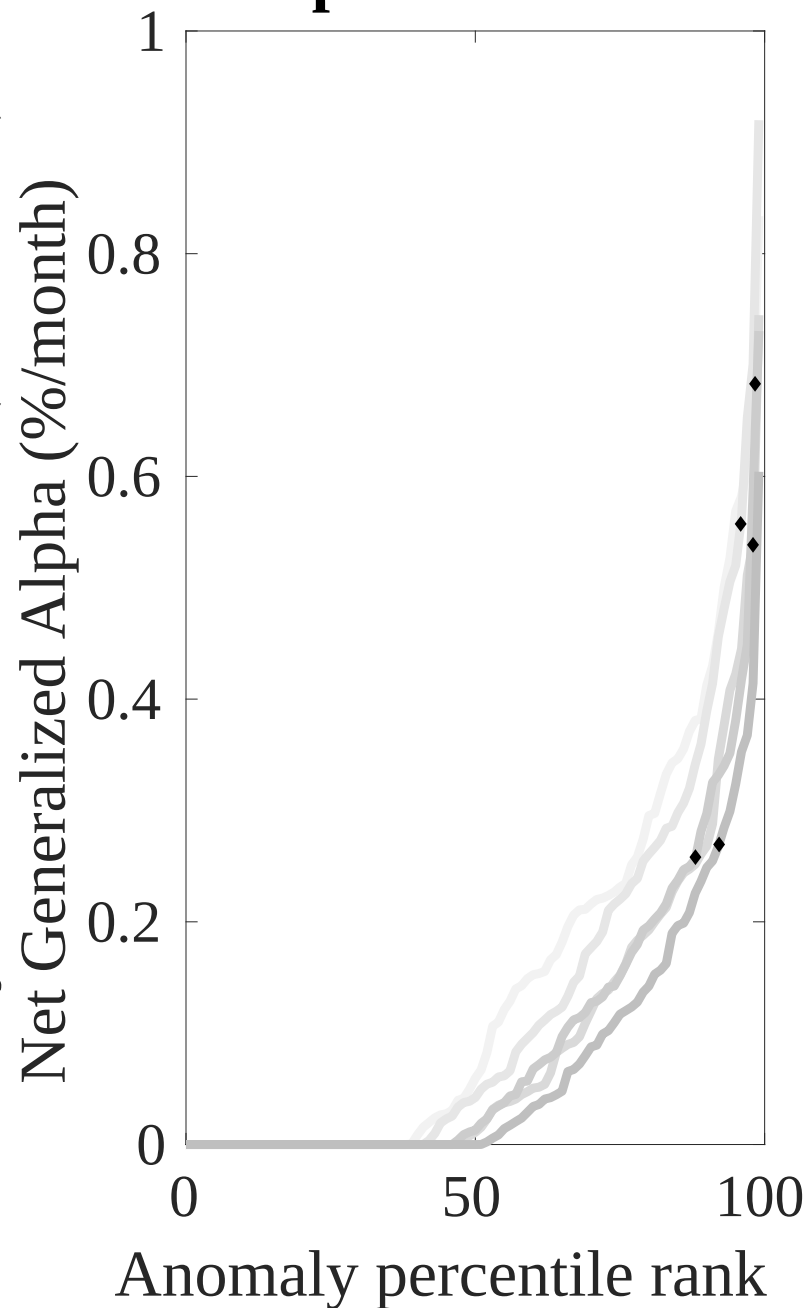
Figure 3: Dollar invested.

This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the TEB trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Net Alpha Percentiles



Novy-Marx and Velikov (FRFS, 2016)

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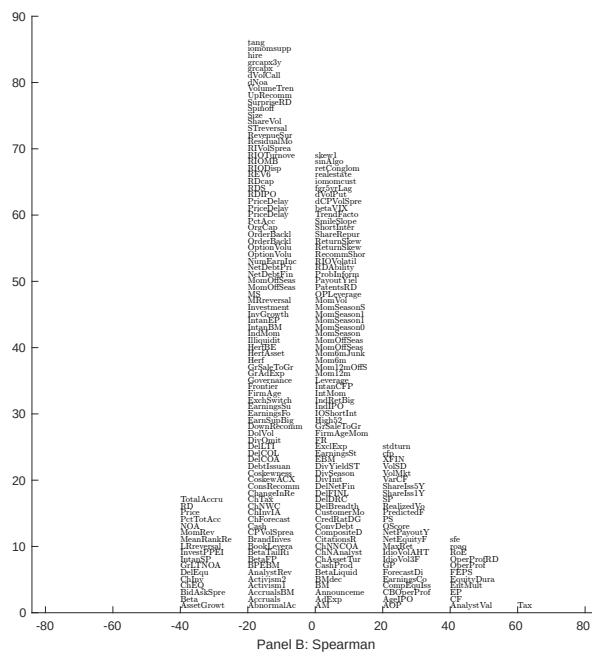
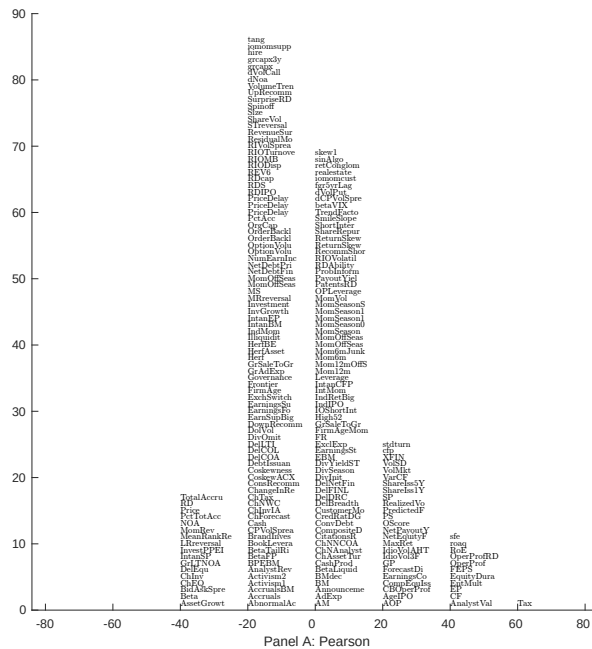


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 209 filtered anomaly signals with TEB. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

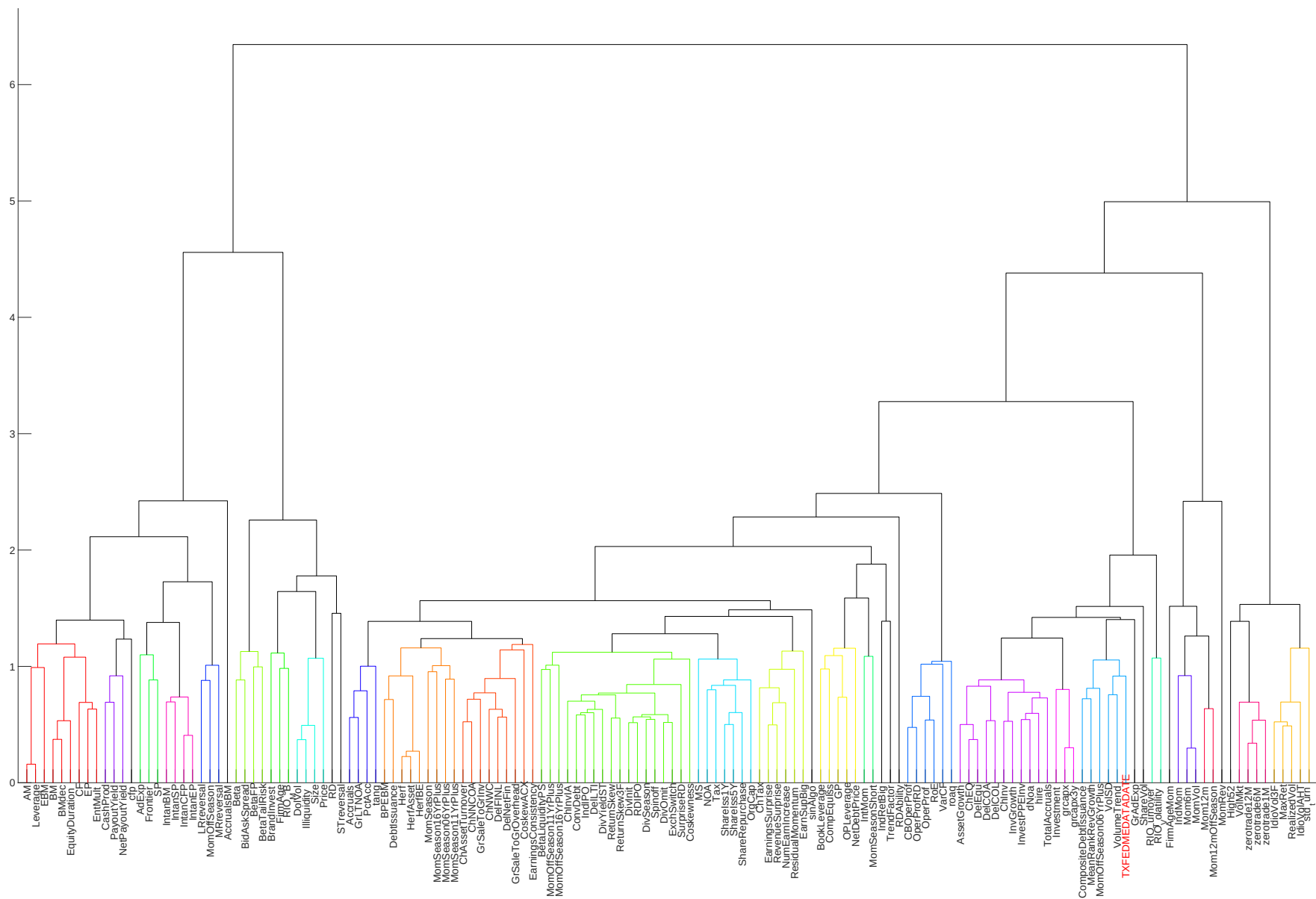


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

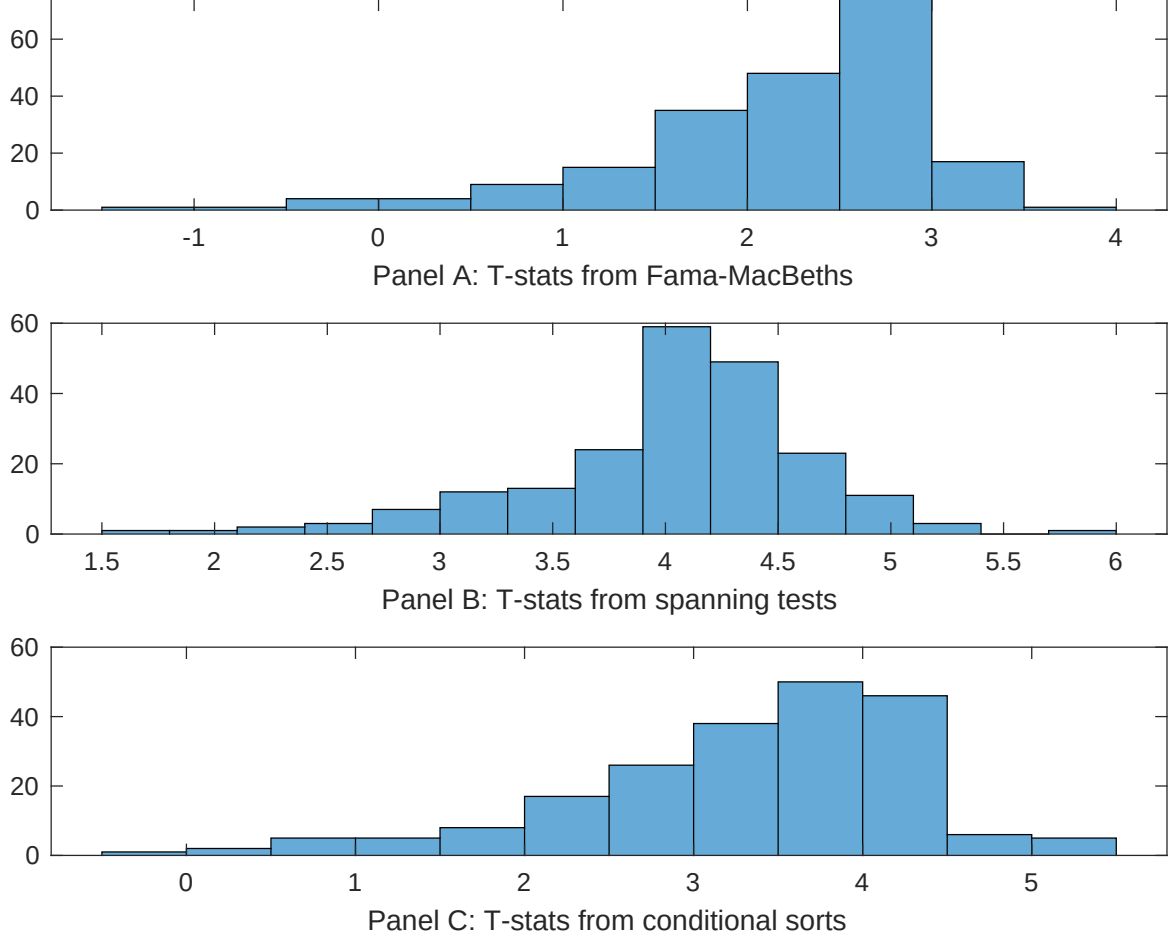


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of TEB conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{TEB} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{TEB}TEB_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{TEB,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on TEB. Stocks are finally grouped into five TEB portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted TEB trading strategies conditioned on each of the 209 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on TEB. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{TEB}TEB_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Analyst earnings per share, Net Payout Yield, EPS Forecast Dispersion, Realized (Total) Volatility, Net equity financing, Idiosyncratic risk (AHT). These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196706 to 202406.

Intercept	0.97 [3.26]	0.12 [4.85]	0.11 [4.36]	0.14 [7.87]	0.13 [5.04]	0.12 [6.65]	0.11 [5.91]
TEB	0.37 [2.52]	0.26 [1.53]	0.50 [2.50]	0.30 [2.65]	0.14 [0.98]	0.31 [3.16]	0.30 [1.30]
Anomaly 1	0.59 [1.09]						-0.20 [-0.08]
Anomaly 2		0.20 [2.11]					0.52 [1.14]
Anomaly 3			0.31 [2.05]				0.30 [3.09]
Anomaly 4				0.14 [3.44]			0.19 [5.51]
Anomaly 5					0.19 [3.25]		0.44 [1.09]
Anomaly 6						0.93 [1.56]	-0.18 [-2.48]
# months	576	668	576	679	629	679	576
$\bar{R}^2(\%)$	2	1	1	2	1	2	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the TEB trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{TEB} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Analyst earnings per share, Net Payout Yield, EPS Forecast Dispersion, Realized (Total) Volatility, Net equity financing, Idiosyncratic risk (AHT). These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196706 to 202406.

Intercept	0.28 [2.70]	0.30 [3.06]	0.25 [2.55]	0.25 [2.47]	0.26 [2.38]	0.24 [2.33]	0.26 [2.68]
Anomaly 1	15.71 [3.76]						-3.31 [-0.73]
Anomaly 2		35.65 [9.39]					29.43 [5.41]
Anomaly 3			29.48 [8.52]				27.32 [6.49]
Anomaly 4				19.42 [6.57]			7.23 [1.74]
Anomaly 5					16.36 [3.35]		-15.01 [-2.37]
Anomaly 6						16.68 [4.79]	-1.41 [-0.29]
mkt	-13.44 [-5.10]	-9.78 [-4.15]	-7.72 [-2.99]	-6.17 [-2.25]	-12.91 [-4.86]	-8.99 [-3.29]	-3.37 [-1.24]
smb	17.62 [3.75]	26.31 [7.70]	19.81 [5.08]	34.30 [8.43]	26.47 [6.26]	36.72 [7.48]	22.82 [4.73]
hml	19.21 [3.97]	14.34 [3.05]	26.92 [5.94]	22.50 [4.81]	27.53 [5.69]	24.05 [5.07]	15.99 [3.27]
rmw	69.41 [10.84]	65.67 [13.10]	59.86 [10.96]	71.25 [13.71]	78.96 [13.73]	71.56 [12.58]	51.26 [8.10]
cma	6.89 [0.97]	-17.89 [-2.46]	2.57 [0.38]	1.37 [0.20]	-1.16 [-0.15]	4.39 [0.62]	-12.88 [-1.76]
umd	1.44 [0.56]	3.65 [1.60]	-0.73 [-0.30]	-3.04 [-1.25]	1.43 [0.57]	-1.58 [-0.65]	0.13 [0.05]
# months	577	680	577	680	630	680	577
$\bar{R}^2(\%)$	55	54	59	51	50	50	62

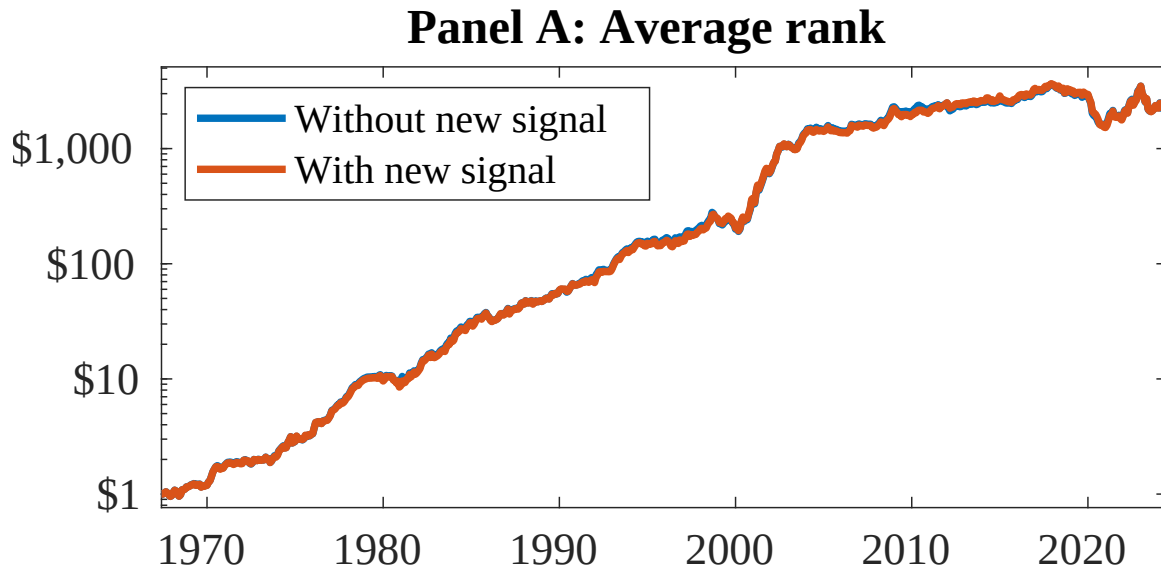


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 156 anomalies. The red solid lines indicate combination trading strategies that utilize the 156 anomalies as well as TEB. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

References

- Bansal, R., Dittmar, R. F., and Lundblad, C. T. (2005). Consumption, dividends, and the cross section of equity returns. *Journal of Finance*, 60(4):1639–1672.
- Bansal, R., Kiku, D., and Yaron, A. (2016). An empirical evaluation of the long-run risks model for asset prices. *Critical Finance Review*, 5(1):183–221.
- Bansal, R. and Yaron, A. (2004). Risks for the long run: A potential resolution of asset pricing puzzles. *Journal of Finance*, 59(4):1481–1509.
- Barro, R. J. (2006). Rare disasters and asset markets in the twentieth century. *Quarterly Journal of Economics*, 121(3):823–866.
- Campbell, J. Y. and Cochrane, J. H. (1999). By force of habit: A consumption-based explanation of aggregate stock market behavior. *Journal of Political Economy*, 107(2):205–251.
- Carhart, M. M. (1997). On persistence in mutual fund performance. *Journal of Finance*, 52:57–82.
- Chen, A. and Velikov, M. (2022). Zeroing in on the expected returns of anomalies. *Journal of Financial and Quantitative Analysis*, Forthcoming.
- Chen, A. Y. and Zimmermann, T. (2022). Open source cross-sectional asset pricing. *Critical Finance Review*, 27(2):207–264.
- Cochrane, J. H., Piazzesi, M., and Tuzel, S. (2023). Consumption-based asset pricing with investment shocks. *Journal of Finance*, 78(2):987–1024.
- Detzel, A., Novy-Marx, R., and Velikov, M. (2022). Model comparison with transaction costs. *Journal of Finance*, Forthcoming.

- Fama, E. F. and French, K. R. (1993). Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, 33(1):3–56.
- Fama, E. F. and French, K. R. (2015). A five-factor asset pricing model. *Journal of Financial Economics*, 116(1):1–22.
- Fama, E. F. and French, K. R. (2018). Choosing factors. *Journal of Financial Economics*, 128(2):234–252.
- Gabaix, X. (2012). Variable rare disasters: An exactly solved framework for ten puzzles in macro-finance. *Quarterly Journal of Economics*, 127(2):645–700.
- Jagannathan, R. and Wang, Y. (2007). Lazy investors, discretionary consumption, and the cross-section of stock returns. *Journal of Finance*, 62(4):1623–1661.
- Kroencke, T. A. and Schindler, F. (2012). Risk premia and consumption fluctuations at different frequencies. *Review of Financial Studies*, 25(12):3841–3871.
- Lettau, M. and Ludvigson, S. (2001). Resurrecting the (c)capm: A cross-sectional test when risk premia are time-varying. *Journal of Political Economy*, 109(6):1238–1287.
- Novy-Marx, R. and Velikov, M. (2016). A taxonomy of anomalies and their trading costs. *Review of Financial Studies*, 29(1):104–147.
- Novy-Marx, R. and Velikov, M. (2023). Assaying anomalies. *Working paper*.
- Parker, J. A. and Julliard, C. (2005). Consumption risk and the cross section of expected returns. *Journal of Political Economy*, 113(1):185–222.
- Santos, T. and Veronesi, P. (2010). Habit formation, the cross section of stock returns and the cash-flow risk puzzle. *Journal of Financial Economics*, 98(2):385–413.

- Savov, A. (2011). The price of skill: Performance evaluation by households. *Journal of Financial Economics*, 102(2):360–378.
- Wachter, J. A. (2013). Can time-varying risk of rare disasters explain aggregate stock market volatility? *Journal of Finance*, 68(3):987–1035.